

Final Exam Solution 2024

Question 1.

- (a) When a dielectric material is subjected to an external electric field, dipoles form at the microscopic level. The process is known as polarization. The dipoles are measured by their moments $\vec{p} = q\vec{d}$, where q is the charge of the dipole and d is the distance by which two opposite polarity charges are separated to form a dipole. The polarization in a dielectric material is thus measured by total dipole moments per unit volume referred as polarization vector.

$$\vec{P} = \frac{\sum q\vec{d}}{\text{Volume}} \quad \text{Cm}^2$$

However, since polarization happens at microscopic level, measurement of $\vec{P}$ in a practical way is not straight forward. Polarization vector $\vec{P}$ is proportional to the external electric field applied.

(ii)
$$\vec{P} = \chi_e (\epsilon_0 \vec{E})$$

~~The polarization vector $\vec{P}$ is also the equal to~~
The polarization causes a bound charge density in the dielectric materials. Hence, it can be concluded that bound charge density $\rho_p = -\vec{\nabla} \cdot \vec{P}$

First postulate in dielectric
$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0} (\rho + \rho_p)$$

Substituting ρ_p ,

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0} (\rho - \vec{\nabla} \cdot \vec{P})$$

$$\vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho$$

Substituting $\vec{P}$

$$\vec{\nabla} \cdot [\epsilon_0 \vec{E} + \chi_e \epsilon_0 \vec{E}] = \rho$$

$$\vec{\nabla} \cdot \epsilon_0 (1 + \chi_e) \vec{E} = \rho$$

$$\vec{\nabla} \cdot \epsilon_0 \epsilon_r \vec{E} = \rho$$

Relative permittivity:
$$\epsilon_r = (1 + \chi_e)$$

(b) (i) Parallel plate capacitor $C = \frac{\epsilon_0 \epsilon_r A}{d} \text{ F}$

$$\epsilon_0 = 8.854 \times 10^{-12}, \quad \epsilon_r = 2.5, \quad A = (35 \times 10^{-3} \times 2.5 \text{ m})^2$$

$$d = 0.1 \times 10^{-3} \text{ m}$$

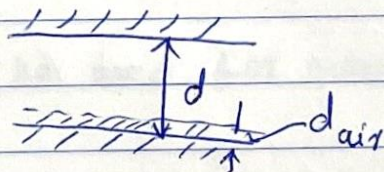
(ii) $E_{\text{max}} = \frac{V_{\text{max}}}{d}$

$$V_{\text{max}} = 20 \times 10^6 \times 0.1 \times 10^{-3} = 2 \text{ kV}$$

(c) 15% of 2 kV = 300 V

With this applied voltage, $E = V/d = 300 / (0.1 \times 10^{-3})$
 $= 3 \text{ MV/m}$

Since, $E = 3 \text{ MV/m}$ is the dielectric strength of air, and it is possible that there is some small tear in the paper, which is this capacitor thus, the capacitor now have paper and air layers.



Assuming, thickness of air layer = d_{air}

" " paper " = $(d - d_{\text{air}}) = d_p$

Two capacitors in series $\frac{1}{C} = \frac{1}{C_{\text{paper}}} + \frac{1}{C_{\text{air}}}$

$$C = \frac{C_p C_a}{C_p + C_a} = \frac{\frac{\epsilon_0 \epsilon_r A}{d_p} \cdot \frac{\epsilon_0 A}{d_a}}{\frac{\epsilon_0 \epsilon_r A}{d_p} + \frac{\epsilon_0 A}{d_a}}$$

$$C = \frac{\frac{(\epsilon_0 \epsilon_1 A) \cancel{\epsilon_0 A}}{d + d_a}}{\frac{(\epsilon_1 d_a + d_p) \cancel{\epsilon_0 A}}{d_p d_a}}$$

$$= \frac{\epsilon_0 \epsilon_1 A}{\epsilon_1 d_a + d_p}$$

if $d_a \rightarrow 0$, $C = \frac{\epsilon_0 \epsilon_1 A}{d}$ (unfaulty capacitor)

Since ~~value~~ ~~is~~ unchanged,

Since after the fault, capacitance value is unchanged, $d_a \rightarrow 0$, $d_p \approx d$ i.e., the tear in the paper is very small.

Question 2

$$(i) \quad C = \frac{\pi \epsilon_0}{\ln \left[\frac{D}{2a} + \sqrt{\left(\frac{D}{2a} \right)^2 - 1} \right]} \text{ F/m} \quad \left\{ \begin{array}{l} D = 4.5 \text{ mm} \\ a = 0.5 \text{ mm} \end{array} \right.$$

Capacitance by Method of image should be used for more accurate value

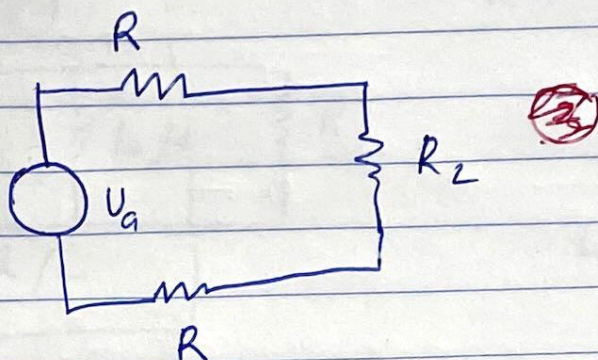
$$C = \frac{\pi \epsilon_0}{\ln \left(\frac{D-a}{a} \right)} \text{ F/m} \quad \left[\text{Half mark, if used} \right]$$

capacitance by Gauss's law, can be used if $D \gg a$ is assumed.

(ii) the resistance of the line conductor, $R = \frac{l}{\sigma S} \Omega$ ①

$$l = 45 \text{ m}, \quad \sigma = 5.8 \times 10^7 \text{ S/m}$$

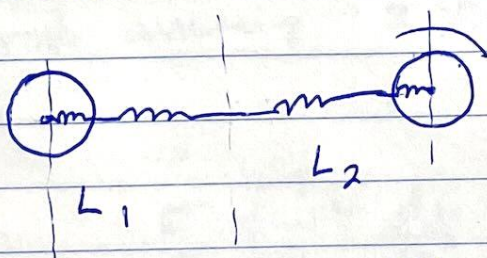
$$S = \pi a^2$$



$$I = \frac{U_g}{2R + R_L} \text{ A} \quad U_R = IR_L \text{ or } U_g - (2R)I$$

①
①

(iii) Inductance between two conductors



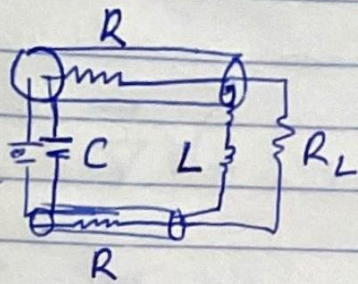
$$L_1 = L_2 = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{d}{a} = L$$

$$L_{\text{Tot}} = L_1 + L_2 = 2L$$

$$L_{\text{Tot}} = \frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln \left(\frac{d}{a} \right) \text{ H/m}$$

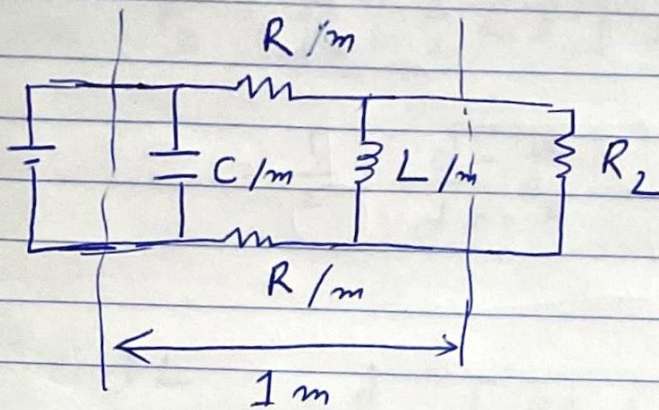
$d = 4.5 \text{ mm}$
 $a = 0.5 \text{ mm}$

(iv)



$$R = \frac{l}{\sigma s} \Omega$$

$$R \text{ per meter} = \frac{1}{\sigma s} \Omega/m$$



C per meter found in (i)

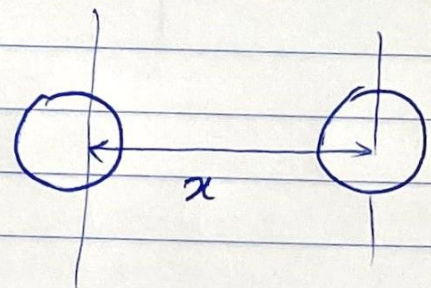
L " " " (iii)

(b) ~~Both~~ Both electrostatic and ~~Electromagnetic~~ force can be found from virtual displacement method

$$\vec{F}_e = \pm \vec{\nabla} W_e, \quad W_e = \frac{1}{2} C V^2$$

Cond assembly $x \gg a$, $C \approx \frac{\pi \epsilon_0}{\ln \frac{x}{a}}$

$$\vec{F}_x = \left[\frac{1}{2} V^2 \frac{dC}{dx} \right] \vec{a}_x$$



$$\frac{dC}{dx} = \frac{d}{dx} \left[\frac{\pi \epsilon_0}{\ln x/a} \right]$$

$$\vec{F}_x = \frac{\pi \epsilon_0}{2} V^2 \frac{1}{x \ln^2(x/a)} \vec{a}_x$$

$$V = 55 \text{ V}$$

$$x = 4.5 \text{ mm}$$

$$a = 0.5 \text{ mm}$$

4 marks

$$(ii) \vec{F}_e = -\vec{\nabla} W_m, \quad W_m = \frac{1}{2} L \dot{\varphi}^2$$

$$\vec{F}_\varphi = \frac{d}{dt} \left[\frac{1}{2} L \dot{\varphi} \right] \vec{e}_\varphi$$

$$= \frac{1}{2} \dot{\varphi}^2 \frac{dL}{d\varphi} \vec{e}_\varphi, \quad L = \frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln\left(\frac{r}{a}\right)$$

$$\frac{dL}{d\varphi} = \frac{d}{d\varphi} \left[\frac{\mu_0}{\pi} \ln\left(\frac{r}{a}\right) \right]$$

$$= \frac{\mu_0}{\pi} \frac{d}{d\varphi} \left[\ln\left(\frac{r}{a}\right) \right] = \frac{\mu_0}{\pi} \frac{1}{r}$$

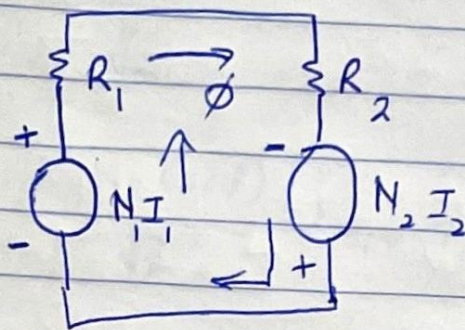
$$\text{so } \frac{dL}{d\varphi}$$

$$\vec{F}_\varphi = \frac{1}{2} \dot{\varphi}^2 \left[\frac{\mu_0}{\pi} \frac{1}{r} \right] \vec{e}_\varphi$$

$$\left. \begin{array}{l} I = \text{current in} \\ a(l/l) \\ r = 4.5 \text{ mm} \end{array} \right\}$$

Question 3

(i)



$$R_1 = \frac{l_1}{\mu_0 s}, \quad R_2 = \frac{l_2}{\mu_0 s}, \quad s = bc$$

(ii)

$$\Phi = \frac{N_1 I_1 - N_2 I_2}{(R_1 + R_2)}$$

$$(iii) \quad B_g = \frac{\Phi}{s} = \frac{1}{s} \cdot \frac{(N_1 I_1 - N_2 I_2)}{\frac{(l_1 + l_2)}{\mu_0 s}}$$

$$= \mu_0 \frac{N_1 I_1 - N_2 I_2}{(l_1 + l_2)}$$

(b) Let's consider $B = B_{max} \sin \omega t$

$$\Phi = B \cdot s = (B_{max} \cdot s) \sin \omega t, \quad \omega = 2\pi f$$

By Faraday's law, $V_2 = -N_2 \frac{d\Phi}{dt}$

$$= -N_2 \frac{d}{dt} (B_{max} s \sin \omega t)$$

$$= (N_2 B_{max} s \omega) \cos \omega t$$

$$V_{2 \text{ rms}} = \frac{N_2 B_{\text{max}} \omega}{\sqrt{2}}$$

$$= \frac{N_2 (2\pi f)}{\sqrt{2}} B_{\text{max}} S_c$$

$$= 4.44 N_2 f B_{\text{max}} S_c$$

(C) $V_1 = 240 \text{ V}$

$$V_1 = 4.44 N_1 f B_{\text{max}} S_c$$

or

For B_{max} to be fixed at 1.1 T, when frequency f changes, V must proportionally change.

$$\frac{V_1}{f} = 4.44 N_1 B_{\text{max}} S_c$$

$$\Rightarrow \frac{V_1}{f} = 4.44 N_1 (B_{\text{max}} S_c) = \frac{240}{50}$$

With new frequency, $4.44 N_1 (B_{\text{max}} S_c)$ will remain unchanged if V_1 is changed proportionally

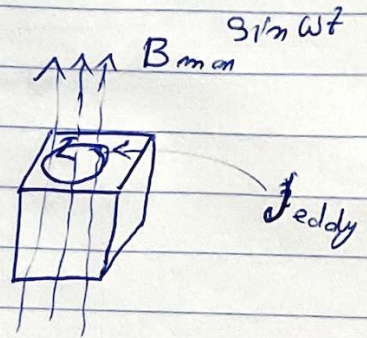
$$\frac{V_1}{f_1} = \frac{240}{50} = \frac{V_{\text{new}}}{16.67 \text{ Hz}}$$

$$V_{\text{new}} = \frac{240}{50} \times 16.67 = 80.02 \text{ V}$$

(d) The time-varying magnetic field of the core induces an emf inside the core, ~~which~~. Because of this emf, a ~~current~~ circulating current induces in the core.

Since the core is conductive, the circulating eddy current will cause Joule's loss ($I_{\text{eddy}}^2 R$), which can heat up the core.

The circulating current also induces a magnetic field that opposes the main magnetic field. This eddy current can also reduce useful magnetic flux in the core.



Eddy current can be reduced by using thin laminated sheet to build the core. The lamination sheets are stacked to form the required depth of the core. Each sheet is electrically insulated from the other, resulting in reduced eddy current and loss.

~~Reduces~~

